

Lidar Introduction and Usage

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1. Lidar Introduction

1.1 T-mini Plus Lidar

The lidar used in this product is the T-mini Plus, a 360° 2D lidar. Based on the pulsed ToF ranging principle, together with associated optical, electrical, and algorithmic design work, the T-mini Plus achieves high-frequency, high-precision distance measurement. At the same time, its mechanical structure rotates 360 degrees to continuously acquire angle information, thereby achieving 360-degree scanning ranging and outputting point cloud data of the scanned environment.



1.2 Lidar Features

- 360-degree all-around scanning, 6-12Hz adjustable scanning frequency
- High-speed ranging, ranging frequency 4000Hz
- Small ranging error, good ranging stability
- Strong resistance to ambient light interference
- Class I eye safety standard

1.3 Performance Parameters

Item	Value	Unit
Ranging frequency	4000	Hz
Scanning frequency	6 (6-12)	Hz
Ranging range	0.05-12	m
Scanning angle	0-360	Deg
Ranging accuracy	20	mm
Angular resolution	0.54	Deg
Pitch angle	0-1.5	Deg